# The Development and Evaluation of a Hybrid ViT-LSTM for Word-Level Sign Language Recognition

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**Abstract.** This paper presents a hybrid Vision Transformer-Long Short-Term Memory (ViT-LSTM) model for word-level sign language recognition (SLR) in low-resource settings. A pretrained ViT backbone extracts per-frame spatial features, which are passed to a Bidirectional LSTM to model temporal dynamics across gesture sequences. A learned attention mechanism produces a context-aware representation for classification. The architecture is evaluated first on MMLAuslan, a large-scale Australian Sign Language (Auslan) dataset, where domain-proximate ViT pretraining achieved 71% validation accuracy, and 55% test accuracy on 35 alphanumeric signs versus 44% for a general-purpose baseline. Trained on a linguistically motivated 116-gloss vocabulary, the model reached 77% validation accuracy and 78% test accuracy using approximately 1,400 training samples. A key contribution is a cross-lingual transfer pipeline from Auslan to BSL, exploiting shared phonological structure between the two languages. BSL fine-tuning uses progressive backbone unfreezing and separate learning rates to preserve transferable representations. Experiments on BSL training using transfer learning show accuracy improvement over a BSL-only baseline. Attention map analysis confirms the model attends to task-relevant hand regions rather than background artefacts. Future work will incorporate pose estimation and hand landmarks, with the longer-term goal of real-time learner feedback for classroom use. The aim is to ensure that SLR development actively supports equitable participation for DHH learners rather than reproducing existing resource inequalities.

## 1 INTRODUCTION

For Deaf and hard-of-hearing (DHH) learners, communication barriers in hearing-centred educational environments remain a persistent challenge. Automatic sign language recognition (SLR) capable of operating in real-time could address this directly, supporting classroom communication assistance, independent vocabulary acquisition, and automated feedback on signing accuracy [5]. Access to such tools is not merely beneficial but a prerequisite for meaningful participation in inclusive education [2]. Yet SLR research has been highly uneven across languages. The vast majority of work targets well-resourced languages such as American Sign Language (ASL), while British Sign Language (BSL) remains largely unsupported by modern deep learning approaches [11, 14]. This imbalance reflects a structural inequality in AI-supported education in which the availability of assistive technology is constrained by the scarcity of large-scale annotated datasets for minority sign languages.

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This paper presents a hybrid Vision Transformer–Long Short-Term Memory (ViT-LSTM) architecture for word-level, isolated SLR, designed explicitly for low-resource settings. Sign languages are spatiotemporal systems [12] in which individual signs are spatially structured within each frame while meaning is also conveyed through movement, timing, and the dynamic evolution of handshape across a temporal sequence [9]. Purely convolutional approaches capture spatial detail but struggle with long-range temporal dependencies [4], while recurrent models handle sequences well but lack strong spatial inductive biases [6]. The proposed hybrid model addresses both challenges within a single architecture.

The system was evaluated in two stages on the MM-WLAuslan dataset [16], a large and balanced Australian Sign Language (Auslan) corpus, before being adapted to a custom BSL dataset via cross-lingual transfer learning. The central contribution is a cross-lingual transfer pipeline from Auslan to BSL, exploiting the phonological proximity of the two languages [1] to compensate for the scarcity of BSL training data.

Section 2 reviews relevant background. Section 3 describes the model architecture and training strategy. Section 4 details the datasets and experimental design. Section 5 reports results. Section 6 discusses the findings, and Section 7 concludes.

## 2 BACKGROUND

### 2.1 Sign Language Recognition

SLR approaches divide broadly into sensor-based and vision-based methods. Sensor-based systems such as datagloves can achieve very low error rates but remain impractical due to cost and scalability constraints [14]. Vision-based deep learning approaches have therefore become dominant. Convolutional Neural Networks (CNNs) have achieved strong results on static sign images [14], but sign language is inherently dynamic. Most signs involve coordinated movement of both hands over time, making purely spatial models insufficient. 3D-CNNs extend to video but lack the global self-attention mechanisms needed to capture long-range spatial dependencies [4].

### 2.2 Vision Transformers

Introduced by Dosovitskiy et al. [4], Vision Transformers (ViTs) apply transformer self-attention directly to sequences of image patches. Unlike CNNs, ViTs integrate information across the entire image simultaneously, making them well suited to the fine-grained spatial structure of sign language, where hand position relative to the face and body carries semantic meaning. A key constraint is that ViTs typically require very large datasets to train from scratch; however, pretrained ViTs transfer effectively to smaller target domains, making them particularly appropriate for low-resource SLR.

## 2.3 Long Short-Term Memory Networks

LSTMs [6] are recurrent networks designed to model long-range temporal dependencies via gating mechanisms. Their application to SLR is well established. Sundar and Bagyammal [18] combined MediaPipe pose extraction with an LSTM to achieve 99% accuracy on ASL fingerspelling, while Huang and Chouvatut [8] coupled a ResNet spatial encoder with an LSTM to obtain 86.25% on Argentine Sign Language. These results confirm that temporal modelling is a critical component of high-performing SLR systems.

## 2.4 Hybrid Architectures and Transfer Learning

Chen et al. [3] demonstrated that combining a ViT feature extractor with a bidirectional LSTM for micro-expression recognition achieved 86.7% accuracy and an F1 of 86.4%, confirming the complementary nature of the two components for spatiotemporal tasks. For low-resource sign languages, Mocialov et al. [15] showed that transfer learning from a related sign language corpus produces better initial performance, faster convergence, and improved final accuracy compared to training from scratch, establishing cross-lingual transfer as a viable strategy for bootstrapping BSL recognition.

# 3 MODEL ARCHITECTURE AND TRAINING STRATEGY

## 3.1 Architecture Overview

The hybrid model comprises three components: a ViT backbone for per-frame spatial feature extraction, a bidirectional LSTM with learned temporal attention for sequence modelling, and a classification head mapping temporal representations to gloss classes. The architecture is illustrated in Figure 1.

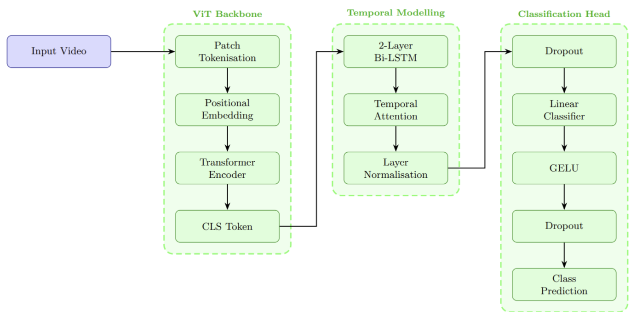


**Figure 1:** Hybrid ViT-LSTM model architecture.

**ViT Backbone.** Each input frame is processed independently by the ViT backbone. The frame is divided into 16×16-pixel patches, linearly projected into token embeddings, and enriched with positional encodings before being passed through the transformer encoder [4]. The resulting CLS token, a 768-dimensional global representation, is used as the per-frame embedding. This compact representation avoids the complexity of pooling the full patch token sequence and provides a stable input to the temporal component. The backbone selected for all experiments was Dima806, a Google ViT pretrained on ImageNet-21k and fine-tuned on HaGRID, a large-scale hand gesture dataset, which outperformed both a general ImageNet backbone and a human action recognition backbone in preliminary experiments.

**Bidirectional LSTM and Temporal Attention.** Sign gestures are dynamic and therefore individual frames are insufficient to discriminate between signs. The 20 per-frame ViT embeddings are passed to a two-layer Bidirectional LSTM (Bi-LSTM), enabling the model to integrate contextual information from both past and future frames within the sequence [3]. A learned scalar attention mechanism is applied over the Bi-LSTM output, assigning importance weights to each frame before producing a context-aware pooled representation. This is preferable to mean pooling because transitional and preparatory frames carry less discriminative information than peak sign frames [10]. The attended representation is layer-normalised before being passed to the classification head.

**Classification Head.** The head consists of two fully connected layers with intermediate dimensionality of 256, a GELU activation between them, and dropout applied before each layer (rates of 0.5 and 0.3 respectively) to regularise training under limited data.

## 3.2 Training Strategy

Training followed a two-phase progressive unfreezing strategy [7, 21]. In Phase 1, the ViT backbone was frozen and only the Bi-LSTM and classification head were updated, protecting pretrained spatial representations from large early gradient updates while allowing the temporal components to stabilise. In Phase 2, the final three transformer layers were selectively unfrozen and updated with a backbone learning rate an order of magnitude lower than the head, with a linear warmup scheduler applied over 10 epochs. Cross-entropy loss with label smoothing of 0.1 was used to prevent overconfidence [19], and AdamW was selected as the optimiser for its improved generalisation in fine-tuning settings [13].

**BSL Transfer Learning Configuration.** For BSL fine-tuning, the best Auslan checkpoint was loaded as the starting weights, with the classification layer reinitialised for the BSL target vocabulary. Phase 2 unfroze eight rather than three backbone layers to improve cross-lingual domain adaptation. Augmentation strength was reduced relative to Auslan training, as the Auslan backbone already encodes substantial visual invariance [17].

Input videos were represented as fixed-length sequences of 20 frames, sampled uniformly with a configurable offset to skip pre-sign pauses, and each frame was resized to 224×224 pixels.

# 4 DATASETS AND EXPERIMENTAL DESIGN

## 4.1 Datasets

**MM-WLAuslan.** The primary dataset was MM-WLAuslan [16], comprising 289,900 videos from 73 signers covering 3,215 Auslan glosses, with 22 videos per gloss providing strong class balance. Only the front-facing RealSense camera was used, reflecting real-world single-camera deployment constraints. Evaluation used the Test-STU subset, which matches the background conditions of the training data, isolating model behaviour from environmental variation.

**Custom BSL Dataset.** No large-scale publicly available dataset exists for isolated BSL recognition [15]. A custom dataset was therefore constructed by scraping and manually segmenting publicly available BSL educational videos from YouTube and dedicated BSL teaching sites, in accordance with approved ethical guidelines. This process introduced naturalistic variability in backgrounds, lighting, and signer style. Each sign was independently verified for accuracy against SignBSL. The resulting dataset is class-imbalanced with a modal class size of 12 videos and some glosses having as few as six training samples.

## 4.2 Vocabulary Selection

Experiments were conducted across two vocabulary stages applied to both Auslan and BSL. Stage 1 comprised alphanumeric signs (letters A–Z and numbers 0–10 if available). Stage 2 expanded to a 116-class Auslan and 85-class BSL vocabulary drawn from an adapted Swadesh list [20], a linguistically motivated selection of high-frequency vocabulary. Dataset statistics are summarised in Table 1.

**Table 1:** Dataset statistics for Auslan and BSL across both vocabulary stages (train/val/test videos).

| Dataset | Stage | Classes | Train | Val/Test |
|---|---|---|---|---|
| Auslan | 1 | 35 | 420 | 70 |
| Auslan | 2 | 116 | 1,392 | 170 |
| BSL | 1 | 36 | 288 | 72 |
| BSL | 2 | 85 | 622 | 170 |

## 4.3 Evaluation Framework

Quantitative evaluation used per-class accuracy, precision, recall, and macro-averaged F1-score, with confusion matrices to analyse misclassification patterns. For BSL, scratch-trained models were compared against transfer-learned models to isolate the contribution of cross-lingual transfer.

To complement test metrics, a qualitative and statistical attention analysis framework was developed. Attention weights from the final ViT transformer layer were extracted and overlaid on representative frames as heatmaps. Four regions of interest (ROIs), Person, Face, Arms, and Hands, were detected automatically using MediaPipe pose and face estimation models, with extrapolation applied to handle occlusion and motion blur. Pre- and post-fine-tuning attention concentration was compared using Wilcoxon signed-rank tests with effect sizes quantified by Cohen's $d$. Spearman's rank correlation was computed between per-ROI attention concentration and classification accuracy to assess whether attention on linguistically relevant regions predicts correct classification. A Bonferroni correction was applied across the four ROIs ($\alpha = 0.0125$).

# 5 RESULTS

## 5.1 Auslan Stage 1 and Stage 2

The full architecture achieved 71.42% validation accuracy on Stage 1 (35-class alphanumeric vocabulary). Scaling to Stage 2 (116-class Swadesh vocabulary) yielded 77.59% validation accuracy, despite a 3.3× increase in class count at the same per-class training density. On unseen testing, 55.71% test accuracy on Stage 1 was achieved. Scaling to Stage 2 improved To 78.45% test accuracy with an F1 of 76.12% (Table 2).

The improvement from Stage 1 to Stage 2 despite the larger vocabulary is attributable to the nature of the signs. Alphanumeric fingerspelling involves subtle static handshapes with minimal inter-class variation, whereas Swadesh lexical signs involve more varied arm movements and body configurations that provide clearer decision boundaries for the temporal modelling component.

**Table 2:** Auslan test performance across vocabulary stages.

| Stage | Acc (%) | Prec (%) | Rec (%) | F1 (%) |
|---|---|---|---|---|
| 1 | 55.71 | 46.86 | 55.71 | 48.35 |
| 2 | 78.45 | 79.01 | 78.45 | 76.12 |

**Attention Analysis.** Table 3 reports Wilcoxon and Spearman results for both Auslan stages. Fine-tuning produced statistically significant increases in attention concentration across all four ROIs at both stages, with effect sizes growing substantially from Stage 1 to Stage 2. The Face ROI reached significance at both stages but with a negligible effect size, consistent with the limited role of non-manual features in isolated lexical recognition.

**Table 3:** Attention analysis for Auslan models. All significant results survive Bonferroni correction ($\alpha = 0.0125$).

| Stage | ROI | $d$ | Wilcoxon | $\rho$ | Corr. |
|---|---|---|---|---|---|
| 1 | Arms | 1.46 | Yes | 0.487 | Yes |
| 1 | Face | 0.36 | Yes | 0.145 | No |
| 1 | Hands | 1.19 | Yes | 0.544 | Yes |
| 1 | Person | 1.87 | Yes | 0.432 | Yes |
| 2 | Arms | 2.02 | Yes | 0.660 | Yes |
| 2 | Face | 0.06 | Yes | 0.100 | No |
| 2 | Hands | 1.52 | Yes | 0.650 | Yes |
| 2 | Person | 2.84 | Yes | 0.650 | Yes |

Spearman correlations were significant for Hands, Arms, and Person at both stages ($\rho = 0.43$–$0.66$), strengthening considerably from Stage 1 to Stage 2. The Arms ROI showed the largest increase ($\rho = 0.487 \rightarrow 0.660$), indicating that temporal arm movement information became increasingly predictive of correct classification as vocabulary complexity grew. Face attention showed no significant correlation at either stage.

**Error Analysis.** Confusion matrix analysis on the Stage 1 test split revealed consistent misclassification between visually similar handshape pairs. K was frequently predicted as D, R as U, and Three as Four, all cases where signs share near-identical spatial configurations distinguishable primarily by subtle wrist orientation or minor finger extension. These confusions are consistent with the lower Stage 1 accuracy relative to Stage 2. Pairs such as N/E and Y/T showed less visually motivated confusions, suggesting the model may require dedicated hand keypoint features to resolve fine-grained handshape distinctions.

## 5.2 BSL Cross-Lingual Transfer

Table 4 reports test performance for all four BSL configurations. Transfer learning from Auslan substantially outperformed training from scratch in both stages.

**Table 4:** BSL test performance: scratch training vs. Auslan-to-BSL transfer.

| Stage | Type | Acc (%) | Prec (%) | Rec (%) | F1 (%) |
|---|---|---|---|---|---|
| 1 | Scratch | 11.11 | 7.54 | 11.11 | 8.32 |
| 1 | Transfer | 25.71 | 22.08 | 25.71 | 21.54 |
| 2 | Scratch | 28.24 | 25.88 | 28.24 | 25.14 |
| 2 | Transfer | 38.82 | 38.98 | 38.82 | 35.79 |

At Stage 1, transfer learning doubled test accuracy relative to the scratch baseline (11.11% to 25.71%, a 14.6 percentage point gain). At Stage 2, the gain was 10.6 percentage points (28.24% to 38.82%). Table 5 reports the corresponding attention statistics.

**Table 5:** Attention analysis for BSL models. Cohen's $d$ reports the pre/post fine-tuning effect size; Spearman $\rho$ reports correlation with classification accuracy. Significance after Bonferroni correction ($\alpha = 0.0125$).

| Stage | Type | ROI | $d$ | Wilcoxon | $\rho$ | Corr. |
|---|---|---|---|---|---|---|
| 1 | Scratch | Arms | 0.00 | No | $-0.16$ | No |
| 1 | Scratch | Hands | 0.00 | No | $-0.13$ | No |
| 1 | Transfer | Arms | $-0.10$ | Yes | 0.03 | No |
| 1 | Transfer | Hands | 0.36 | Yes | 0.15 | No |
| 1 | Transfer | Person | 0.24 | Yes | 0.10 | No |
| 2 | Scratch | Arms | 0.51 | Yes | 0.04 | No |
| 2 | Scratch | Hands | 0.60 | Yes | 0.02 | No |
| 2 | Scratch | Person | 0.79 | Yes | 0.03 | No |
| 2 | Transfer | Arms | 0.10 | Yes | $-0.06$ | No |
| 2 | Transfer | Hands | 0.62 | Yes | 0.11 | No |
| 2 | Transfer | Person | 0.35 | Yes | $-0.02$ | No |

The Stage 1 scratch model showed no measurable change in attention concentration following fine-tuning (all $d = 0.0$), indicating that the unfrozen backbone received no effective gradient signal under extreme data scarcity. The Stage 1 transfer model showed significant shifts in Hands and Person attention, confirming that the Auslan initialisation provided a stable enough foundation for spatial learning to proceed. Figure 2 illustrates this contrast: the transfer model develops clear hand-focused attention after fine-tuning, while the scratch model produces no comparable shift.


**(a)** Scratch Model

**Figure 2:** Temporal attention visualisation for BSL Stage 2 Scratch vs Transfer after training.

Spearman correlations between attention concentration and classification accuracy were not statistically significant in any BSL configuration, in marked contrast to the Auslan results. Auslan models were trained and evaluated on the same signer pool with consistent studio backgrounds, enabling stable correspondences between spatial attention and linguistic content. BSL models faced naturalistic recording variability combined with as few as eight training samples per class, insufficient to establish reliable attention-accuracy relationships even when spatial attention did shift toward relevant regions.


**(b)** Transfer Model

**Figure 2:** Temporal attention visualisation for BSL Stage 2 Scratch vs Transfer after training (continued).

# 6 DISCUSSION

## 6.1 Scalability Across Vocabulary Complexity

The improvement from Stage 1 to Stage 2 despite a 3.3× class count increase reflects a fundamental property of sign language vocabulary rather than a limitation of the Stage 1 task. Alphanumeric fingerspelling compresses 35 classes into a highly confusable handshape space in which many letters differ only in subtle wrist rotation, finger extension, or brief movement. The Swadesh lexical signs, by contrast, involve full arm trajectories, body locations, and temporal dynamics that create well-separated decision boundaries.

The Spearman correlation data quantifies this shift precisely. The Arms ROI correlation with classification accuracy increased from $\rho = 0.487$ at Stage 1 to $\rho = 0.660$ at Stage 2, indicating that the Bi-LSTM's temporal modelling of arm movement became substantially more predictive of correct classification as vocabulary complexity grew. This is precisely the computational advantage of the hybrid architecture over static image approaches: arm movement over time is the modality that separates lexical signs most clearly, and it is only captured by the temporal component. The Hands correlation also strengthened ($\rho = 0.544 \rightarrow 0.650$), while the Face ROI remained non-significant at both stages. These results suggest the hybrid architecture's value increases with vocabulary complexity, a practically important property for any system intended to support classroom communication rather than laboratory fingerspelling.

## 6.2 Cross-Lingual Transfer from Auslan to BSL

The Stage 1 scratch model showed no measurable change in spatial attention following fine-tuning (all $d = 0.0$), indicating that the unfrozen backbone received insufficient gradient signal to update its representations at all. Training from scratch on 288 samples across 36 classes did not provide enough signal to shift a randomly initialised backbone. The transfer model, starting from Auslan-adapted representations, had enough spatial prior to begin learning under the same data conditions. This is consistent with Mocialov et al.'s finding [15] that cross-lingual transfer produces better initial performance even when source and target languages are structurally distinct.

At Stage 2, the transfer benefit narrowed, with scratch and transfer models converging to 25.14% and 35.79% F1 respectively. This convergence pattern is consistent with data scarcity becoming the dominant constraint at higher vocabulary complexity. With 85 classes and an average of seven to eight training samples per class, neither initialisation strategy can overcome the fundamental limitation of insufficient labelled data. The architecture and pretraining are not the bottleneck at Stage 2: the BSL dataset is.

### 6.3 Attention Analysis as a Reusable Evaluation Framework

The attention analysis framework developed here provides statistically grounded evidence of whether a model is learning linguistically motivated representations, which is a methodological contribution applicable to future spatiotemporal SLR research. The divergence between Auslan and BSL Spearman results illustrates the framework's diagnostic value. Both model families showed significant attention shifts toward hands and arms following fine-tuning. Yet only the Auslan models showed significant correlations between that attention and correct classification. This distinction reveals that attending to the right region and reliably using that region for classification are separable phenomena. The BSL models learned to look in approximately the right places but lacked sufficient data to convert spatial attention into reliable discriminative signal. A framework measuring only attention maps or only classification accuracy would have missed this distinction entirely.

### 6.4 Educational Implications and Limitations

The results have direct implications for the development of assistive technology in educational settings. Over 90% of Deaf children are born to hearing, non-signing families, placing them at acute risk of language deprivation [12]. An SLR system capable of providing real-time signing feedback in classroom and home environments could meaningfully reduce this risk, supporting early language exposure and independent vocabulary acquisition. The Swadesh vocabulary evaluated in Stage 2 was selected precisely because it reflects high-frequency communicative vocabulary relevant to everyday educational interaction rather than laboratory sign sets. The 78.45% accuracy achieved on this vocabulary, under low-resource constraints with approximately 12 training samples per class, demonstrates that the architecture is viable for practical deployment in educational assistive tools without requiring large institutional data collection efforts.

The most significant limitation is BSL dataset scale. With a small number of training samples per class drawn from the same small pool of signers, models are vulnerable to learning signer-specific features rather than linguistically general ones. True signer-independent evaluation would be an essential step before deployment. Covariate shift between clean Auslan studio recordings and naturalistic BSL educational videos also limits transferability, and likely accounts for a meaningful fraction of the performance gap between Auslan and BSL results. Future work should prioritise dedicated BSL data collection, few-shot learning approaches [15], and extension toward real-time classroom feedback tools that could provide DHH learners and their families with accessible, low-cost signing support.

## 7 CONCLUSION

This paper presented a hybrid ViT-LSTM architecture for low-resource word-level sign language recognition, motivated by the need for accessible assistive tools for DHH learners in educational environments. The system was evaluated across two vocabulary stages on Auslan and via cross-lingual transfer to BSL. Three findings emerged. First, the architecture scales effectively with vocabulary complexity: Stage 2 achieved 78.45% test accuracy on 116 classes and outperformed Stage 1 on 35 classes, attributed to the greater temporal distinctiveness of lexical signs over fingerspelling. Second, cross-lingual transfer from Auslan to BSL provides meaningful gains under extreme data scarcity, doubling test accuracy at Stage 1 despite structural differences between the two languages' fingerspelling systems. Third, the attention analysis framework developed here provides interpretable and statistically grounded evidence of linguistically motivated spatial learning, with Spearman correlations confirming that Hands and Arms attention predicts correct classification in Auslan models while simultaneously revealing that data scarcity prevents this relationship from forming in BSL models.

Future work should prioritise larger BSL dataset construction, signer-independent evaluation, and incorporation of explicit hand keypoint features. Extension to continuous SLR and classroom learner feedback tools would realise the educational motivation of this work, contributing toward equitable AI-supported education for DHH communities.

## 8 *

REFERENCES


[1] 'Lexical Comparison of Signs From American, Australian, British, and New Zealand Sign Languages: David McKee and Graeme Kennedy', in *The Signs of Language Revisited*, Psychology Press, (2000).

[2] Ben Ambridge, 'A Computational Simulation of Children's Language Acquisition (Crazy New Idea)', *OASIcs, Volume 93, LDK 2021*, **93**, 4:1–4:3, (2021). https://drops.dagstuhl.de/entities/document/10.4230/OASIcs.LDK.2021.4.

[3] Huilong Chen, Jianjiang Cui, Yucheng Zhang, and Yonglin Zhang, 'VIT and Bi-LSTM for Micro-Expressions Recognition', in *2022 IEEE 5th International Conference on Information Systems and Computer Aided Education (ICISCAE)*, pp. 946–951, (September 2022). https://ieeexplore.ieee.org/document/9927522.

[4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. http://arxiv.org/abs/2010.11929, June 2021.

[5] Faye Ginsburg, 'Disability in the Digital Age', in *Digital Anthropology*, Routledge, (2012).

[6] Sepp Hochreiter and Jürgen Schmidhuber, 'Long Short-Term Memory', *Neural Computation*, **9**(8), 1735–1780, (November 1997).

[7] Jeremy Howard and Sebastian Ruder. Universal Language Model Fine-tuning for Text Classification. http://arxiv.org/abs/1801.06146, May 2018.

[8] Jiayu Huang and Varin Chouvatut, 'Video-Based Sign Language Recognition via ResNet and LSTM Network', *Journal of Imaging*, **10**(6), 149, (June 2024). https://www.mdpi.com/2313-433X/10/6/149.

[9] Tisha Jantzen, 'The Power of Sign: Enhancing Oral Communication with Young Children with Typical Hearing', *Research Papers*, (January 2011). https://opensiuc.lib.siu.edu/gs_rp/171.

[10] Zhenni Li, Yujie Li, Benying Tan, Shuxue Ding, and Shengli Xie, 'Structured Sparse Coding With the Group Log-regularizer for Key Frame Extraction', *IEEE/CAA Journal of Automatica Sinica*, **9**(10), 1818–1830, (October 2022). https://ieeexplore.ieee.org/document/9774986.

[11] Zeyu Liang, Huailing Li, and Jianping Chai, 'Sign Language Translation: A Survey of Approaches and Techniques', *Electronics*, **12**(12), 2678, (June 2023). https://www.mdpi.com/2079-9292/12/12/2678.

[12] Diane Lillo-Martin and Jonathan Henner, 'Acquisition of Sign Languages', *Annual Review of Linguistics*, **7**(1), 395–419, (January 2021). https://pmc.ncbi.nlm.nih.gov/articles/PMC8570554/.

[13] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. http://arxiv.org/abs/1711.05101, January 2019.

[14] M. Madhiarasan and Partha Pratim Roy. A Comprehensive Review of Sign Language Recognition: Different Types, Modalities, and Datasets. http://arxiv.org/abs/2204.03328, April 2022.

[15] Boris Mocialov, Graham Turner, and Helen Hastie. Transfer Learning for British Sign Language Modelling. http://arxiv.org/abs/2006.02144, June 2020.

[16] Xin Shen, Heming Du, Hongwei Sheng, Shuyun Wang, Hui Chen, Huiqiang Chen, Zhuojie Wu, Xiaobiao Du, Jiaying Ying, Ruihan Lu, Qingzheng Xu, and Xin Yu. MM-WLAuslan: Multi-View Multi-Modal Word-Level Australian Sign Language Recognition Dataset. http://arxiv.org/abs/2410.19488, October 2024.

[17] Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your ViT? Data, Augmentation, and Regularization in Vision Transformers. http://arxiv.org/abs/2106.10270, June 2022.

[18] B Sundar and T Bagyammal, 'American Sign Language Recognition for Alphabets Using MediaPipe and LSTM', *Procedia Computer Science*, **215**, 642–651, (2022). https://www.sciencedirect.com/science/article/pii/S1877050922021378.

[19] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the Inception Architecture for Computer Vision. http://arxiv.org/abs/1512.00567, December 2015.

[20] James Woodward, 'Sign languages and sign language families in Thailand and Viet Nam', *The Signs of Language Revisited: An Anthology in Honor of Ursula Bellugi and Edward Klima*, (January 2000).

[21] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? http://arxiv.org/abs/1411.1792, November 2014.